

Compact Generative Models of Human Interception Movements

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Abstract

Human interception requires coordinating movement with a moving target, yet repeated attempts vary within and between people. We investigate whether a compact representation of a participant can support a model of this variability. Using 4,732 free-eye-movement trials from 28 participants, we compared variational autoencoders (VAE), conditional VAE (CVAE), conditional autoencoders (CAE), and spline coefficients compressed by principal component analysis (PCA), with three or eight latent coordinates. Four participant-held-out rotations evaluated trajectory reconstruction and generation from separate context trials. Spline + PCA achieved the lowest reconstruction error. For generation at eight dimensions, VAE and CVAE improved on spline + PCA in both kinematic-feature and full-trajectory distribution comparisons. Their full-trajectory energy discrepancies were 0.0548 and 0.0549, compared with 0.0981 for spline + PCA; their difference was not significant. Replacing a participant's own latent center with population or other-participant centers worsened feature fidelity. However, directly resampling stored context trajectories achieved a lower mean full-trajectory discrepancy than the compact models. We provide an interactive dashboard, with VAE at eight dimensions as the default, and distinguish useful session-specific compression from a complete account of individual movement variability.

1 Introduction

Intercepting a moving object requires predicting where and when the object will be reached while adjusting an ongoing movement. Repeated movements can differ in their onset, duration, curvature and speed profile, even within one participant. Modeling only an average trajectory omits this variability. Our practical objective is therefore to represent distributions of interception movements, including both their detailed paths and interpretable kinematic features, rather than to predict a single deterministic response.

A compact participant representation would make such a model easier to inspect and manipulate. We ask whether averaging a participant's encoded context trials into a small vector, termed a fingerprint, provides useful information for generating the remainder of that participant's session. Three coordinates address the original compact-representation objective, while eight provide a comparison with greater capacity. This definition makes the fingerprint an operational summary of recordings; it does not assume that its individual coordinates correspond to named movement strategies.

Variational autoencoders learn nonlinear latent representations with a probabilistic training objective [1]. Conditional variants additionally receive observed task variables [2]. We compare these approaches with a deterministic neural autoencoder and a statistical pipeline that fits smooth spline coefficients before applying PCA. This comparison separates the value of a nonlinear decoder, probabilistic regularization and explicit task conditioning. Reconstruction first tests how much of an observed trajectory survives compression. Generation then tests whether a participant's context representation produces appropriate variability on separate trials. A subsequent full-trajectory analysis complements feature-based distances by preserving the ordered path.

The project deliverable is a dashboard that exposes the evaluated models, generated movements and benchmark results. Model selection follows the intended generative use: a model can reproduce an observed path accurately yet generate an inadequate distribution. We therefore report both successes and the stronger empirical reference obtained by retaining and resampling context trajectories. Additional pipeline details, corrected comparisons and secondary investigations appear in the accompanying appendix.

2 Data and preprocessing

Jason Friedman's laboratory supplied finger-tracking recordings from an interception task. Participants moved from a starting region to a fixed interception region while a circular target approached horizontally (Figure 1). The target started at one of three categories on either side. Each category had its own speed range, so starting position and speed were related by the experimental design. We retained the free-eye-movement condition (condition 2). The five model condition values were a three-element one-hot starting-category vector, a side indicator and executed target speed normalized by screen width. Participant identity and trial outcomes were not condition inputs.

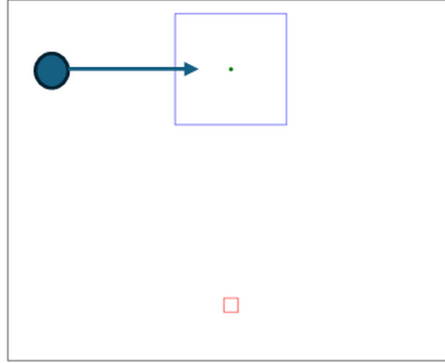


Figure 1. Task illustration adapted in the project presentation from the supplied display: lower starting region, upper interception region and horizontally approaching target. The display illustrates the task, not the tracking apparatus.

Of 4,763 available trials from 28 participants, four lacked an arrival event and 27 arrived more than one second after the target window ended. The resulting 4,732 trials included early arrivals and fixation warnings. Twenty-six participants contributed 171-180 trials; the remaining two contributed 60 and 28 from 60 and 29 available recordings, respectively. The median retained count was 179.5. Each of four fixed rotations used 17 training, four validation and seven test participants, and tested each participant once. All learned transformations used training data; validation data selected settings and checkpoints.

Finger positions were recorded in three dimensions at 240 Hz. Duplicate frame counters were averaged and missing counters linearly interpolated onto a regular grid. A fourth-order 10 Hz Butterworth filter applied forward and backward smoothed positions. The paired target recordings identified target-motion onset. We modeled the interval from that event to the last tracker sample, retaining waiting and movement within this window. The last sample was an arrival proxy, on average 26.2 ms before the MATLAB arrival event. The earlier target-appearance interval was outside this modeled window.

Trials had different durations and sample counts. For a filtered window of L samples, the original phases were $i/(L - 1)$. A cubic interpolant S evaluated at $j/99$ produced 100 points, translated to start at zero:

$$\mathbf{P}_j = \mathbf{S}(j/99) - \mathbf{S}(0), \quad j = 0, \dots, 99.$$

The planar x-y coordinates formed a 100×2 array, flattened in time order to 200 neural inputs. This resampling preserves relative temporal progression; it does not make durations equal in seconds or scale spatial amplitude. Initiation and movement durations were separate supervised targets, transformed as $\log(t + 0.001)$ and standardized on training trials. Onset used a sustained three-frame crossing of 5 tracker units/s in 3-D speed. There were 357 zero-initiation labels (7.5%), including movements already above threshold at target motion. These movements remained in the analysis. Tracker units and event assumptions awaiting laboratory confirmation are listed in Appendix A1.

3 Compact trajectory models

3.1 Spline coefficients, PCA and timing regression

The statistical model first expressed each resampled coordinate as a cubic B-spline. Five interior knots at 1/6, ..., 5/6 produced nine basis functions per axis and a common design matrix B of size 100×9 . Separate least-squares fits gave nine x coefficients and nine y coefficients:

$$\mathbf{c}_x = \arg \min_{\mathbf{c}} \|\mathbf{x} - B\mathbf{c}\|_2^2, \quad \mathbf{c}_y = \arg \min_{\mathbf{c}} \|\mathbf{y} - B\mathbf{c}\|_2^2.$$

Concatenating these coefficients yielded one 18-value row per trial. Stacking only training rows formed the $N_{\text{train}} \times 18$ matrix on which PCA was fitted. Thus PCA compressed spline coefficients, rather than the 100 sampled points or individual basis functions. Validation reconstruction MSE selected between raw and standardized coefficient variants. Let $\mathbf{m}$ and $\mathbf{d}$ denote the effective centering and scaling vectors ($d = 1$ for raw coefficients), and let V contain the n retained PCA directions:

$$\mathbf{z} = V^T[(\mathbf{c} - \mathbf{m}) \oslash \mathbf{d}], \quad \hat{\mathbf{c}} = \mathbf{m} + \mathbf{d} \odot (V\mathbf{z}).$$

Decoding split the reconstructed coefficient vector into two nine-value vectors and multiplied each by B , restoring 100×2 positions. A separate Ridge regression predicted two standardized log-duration targets from the n scores and five task conditions. Training-only input scaling and validation-selected regularization completed this timing head. Conditions therefore affected spline timing, but not its spatial decoder. Appendix A2 gives the matrix dimensions and inverse transformations.

3.2 Variational and deterministic neural models

Neural models standardized each of the 200 coordinate inputs using training-trial statistics. Encoder and decoder each used two 256-unit ReLU hidden layers. A variational encoder predicted an n -dimensional mean and log variance, with training samples obtained by the reparameterization $\mathbf{z} = \mu + \sigma$ multiplied elementwise by standard-normal noise [1]. The decoder predicted 200 standardized coordinates and two standardized log durations. Its per-trial objective was:

$$\mathcal{L} = \|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + 20\|\mathbf{y} - \hat{\mathbf{y}}\|_2^2 + \beta D_{\text{KL}}[q(\mathbf{z} | \mathbf{v}, \mathbf{c}) \| \mathcal{N}(\mathbf{0}, I)].$$

Here $\mathbf{v}$ denotes the standardized trajectory, $\mathbf{y}$ the two standardized timing targets and $\mathbf{c}$ the task conditions. Both squared-error terms are *sums* over coordinates, followed by a batch mean; replacing them with separately averaged MSEs would change their relative weight. The KL term regularizes the posterior toward a standard normal, with beta increased from zero to one over 50 epochs. The conditional VAE supplied the five conditions to both encoder and decoder. Our VAE used the same framework with these inputs masked to zero. The CAE used a deterministic code and omitted the KL term.

Training used Adam (learning rate 0.001), batches of 64, inverse-participant-trial-count sampling, gradient clipping and validation-based learning-rate reduction and early stopping. Three seeds (42, 43, 44) were evaluated per neural model and rotation. The matched comparisons in this report use $n = 3$ and $n = 8$. Reconstruction decodes the observed trial's latent mean or deterministic code; it receives the trial trajectory but not its physical timing targets. The condition-only Ridge comparator receives task conditions without an observed trajectory or personal code. Appendix A3 provides the exact training and inference stages.

4 Generation and statistical evaluation

To evaluate a previously unseen participant, we split that person's trials approximately in half into context and query sets, stratified by starting category and side. The split was shared across models. Only context trajectories formed the participant fingerprint: their encoded means were averaged to an n-value center. We sampled 120 latent vectors around that center using a covariance estimated from training participants:

$$\theta_s = \frac{1}{|C_s|} \sum_{i \in C_s} \mu_i, \quad \mathbf{z}^{(b)} \sim \mathcal{N}(\theta_s, \Sigma_{\text{train}}).$$

The shared covariance combined within-participant variation of training codes with mean posterior variance for variational models and a small diagonal regularizer. CAE and spline omitted posterior variance. This is a participant-specific center with a shared noise model, not a separately estimated covariance for every participant. Generation used the query task-condition mixture where applicable, but never query trajectories or timing labels. The VAE ignored those conditions. Appendix A4 specifies the covariance and sampling procedure.

4.1 Reconstruction and feature distributions

Reconstruction error was the mean squared coordinate difference across 100 x 2 positions, in squared tracker units. For generation, eleven features described timing, speed, path geometry, endpoints and a heuristic speed-peak count (Appendix A5). Recorded features used movement-onset-to-end paths. Generated features used predicted initiation and movement durations to crop the generated path and recover a time scale. Thus feature fidelity also depended on timing predictions.

For each participant and feature, the Kolmogorov-Smirnov (KS) statistic measured the largest difference between recorded and generated empirical cumulative distributions. We averaged the eleven KS statistics. Joint feature comparisons used energy discrepancy [3] and squared maximum mean discrepancy (MMD) with a Gaussian kernel [4], after training-derived feature standardization. KS values here are discrepancies, not hypothesis-test p-values.

4.2 Complete trajectory distributions

The follow-up analysis compared entire generated and query paths without additional cropping or filtering. Flattening a 100 x 2 path and dividing by the square root of 200 made Euclidean distance equal to root mean squared coordinate difference. Energy discrepancy compared cross-distribution distances with within-distribution variation:

$$E = 2 \overline{\|X - Y\|} - \overline{\|X - X'\|} - \overline{\|Y - Y'\|}.$$

Within-sample means included diagonal zeros for energy. MMD used a Gaussian kernel, with bandwidth fixed from training paths and within-sample diagonals excluded. An axis-balanced sensitivity analysis divided each coordinate axis by its training RMS variation. Lower discrepancies indicate closer distributions. They are not accuracy percentages. Repeating the context mean and resampling stored context paths provided additional diagnostic references.

4.3 Aggregation and comparisons

We averaged neural seeds within each participant and weighted all 28 participants equally. Paired two-sided Pratt Wilcoxon tests compared participant scores, with differences rounded to 12 decimals. Benjamini-Hochberg (BH) correction [5] covered the original 140 model comparisons (10 pairs, two dimensions, seven endpoints). The new trajectory analysis used a separate family of 80 comparisons (10 pairs, two dimensions, two metrics, two geometries). We also report Holm sensitivity. These are exploratory comparisons on overlapping training rotations, rather than independent replications. Neither the capacity comparisons nor the context-resampling reference received new significance tests. Appendix A6 gives estimators, assumptions and corrected values.

5 Reconstruction and generated kinematic features

Reconstruction established the fidelity of each compressed representation before testing generation. Spline + PCA had the lowest cohort-average trajectory MSE at both capacities (Table 1). Increasing capacity reduced all four models' mean errors descriptively. The best neural reconstruction differed by capacity: VAE at three dimensions and CAE at eight. Therefore a single reconstruction ranking does not determine the generative model selected for the dashboard.

Table 1. Participant-averaged reconstruction MSE; lower is better. Spline + PCA beat each neural model at each dimension under BH correction over 140 tests (all $q < 0.0031$). The spline-versus-CAE $n = 8$ comparison did not survive Holm correction ($p = 0.106$); the other five did. Within neural models, VAE $n = 3$ beat CAE and CVAE under BH, while CAE $n = 8$ beat both variational models. Condition-only Ridge MSE was 1.233795.

Model	$n = 3$ MSE	$n = 8$ MSE
Spline + PCA	0.097255	0.022772
CAE	0.169661	0.025968
CVAE	0.176981	0.032481
VAE	0.148299	0.033487

Generation showed a different pattern. VAE and CVAE produced smaller feature discrepancies than spline + PCA at both capacities (Table 2). At eight dimensions, VAE had the lowest numerical mean for all three reported feature metrics. Compared with CVAE, however, its energy advantage was significant only under BH ($q = 0.0257$), not Holm; KS ($q = 0.137$) and MMD ($q = 0.0518$) did not cross the BH threshold. These results support VAE as a practical default without establishing a uniquely superior variational architecture.

Table 2. Generation fidelity on eleven kinematic features; all entries are participant means and lower is better. VAE and CVAE each beat spline + PCA on all three metrics at $n = 3$ and $n = 8$ under both BH and Holm correction over 140 tests. CAE $n = 8$ did not differ significantly from spline on these metrics. Feature energy/MMD use standardized eleven-feature vectors and cannot be numerically compared with the trajectory distances in Table 3. Full selected comparisons appear in Appendix A7.

Model	n	Mean KS	Energy	MMD squared
Spline + PCA	3	0.3251	0.9774	0.1815
CAE	3	0.2876	0.8015	0.1414
CVAE	3	0.2778	0.7454	0.1309
VAE	3	0.2846	0.7476	0.1303
Spline + PCA	8	0.2548	0.7347	0.1153
CAE	8	0.2587	0.6202	0.1111
CVAE	8	0.2194	0.4305	0.0675
VAE	8	0.2151	0.4110	0.0639

Among compact models at $n = 3$, CVAE had the lowest mean KS (0.2778), whereas VAE had the lowest joint MMD (0.1303). This metric dependence reinforces the need to state the endpoint and capacity when naming a winner. The condition-only Ridge reference had mean KS 0.2773, energy 0.8187 and MMD 0.1259. It was included in the original correction family and remains available in Appendix A7. Its comparatively plausible feature summaries, despite poor reconstruction, also motivate comparing whole paths.

The feature measures address complementary properties. Mean KS evaluates each marginal distribution separately, whereas energy and MMD assess the joint eleven-dimensional distribution. Correlated features can place extra emphasis on related properties in the joint distances. We retained the specified feature set for this analysis; reducing it after inspecting results would define a new analysis rather than repair the existing p-values. No per-feature KS p-values were combined into the reported mean KS.

6 Full-trajectory generation

A generator could match selected movement features while missing details of the path. We therefore evaluated the complete ordered 100-point trajectories from the same frozen generations. At eight dimensions, VAE and CVAE both improved on spline + PCA in energy and MMD (Table 3). Their raw-trajectory energy means were almost identical. This extends the feature result to the distribution of phase-normalized paths, while leaving physical-time fidelity a separate question.

Table 3. Full-trajectory distribution discrepancies in the primary RMS geometry. At $n = 8$, VAE and CVAE each beat spline + PCA on both metrics under BH and Holm correction across all 80 tests. VAE versus CVAE was not significant (BH $q = 0.672$ for energy and 1.000 for MMD). At $n = 3$, each variational model beat spline on MMD (BH $q = 0.00184$, Holm $p = 0.0363$), but not energy (VAE $q = 0.0647$; CVAE $q = 0.205$). Lower is better.

Model	n	Trajectory energy	Trajectory MMD squared
Spline + PCA	3	0.1068	0.0777
CAE	3	0.1272	0.0818
CVAE	3	0.1024	0.0578
VAE	3	0.1005	0.0572
Spline + PCA	8	0.0981	0.0705
CAE	8	0.0947	0.0652
CVAE	8	0.0549	0.0244
VAE	8	0.0548	0.0238

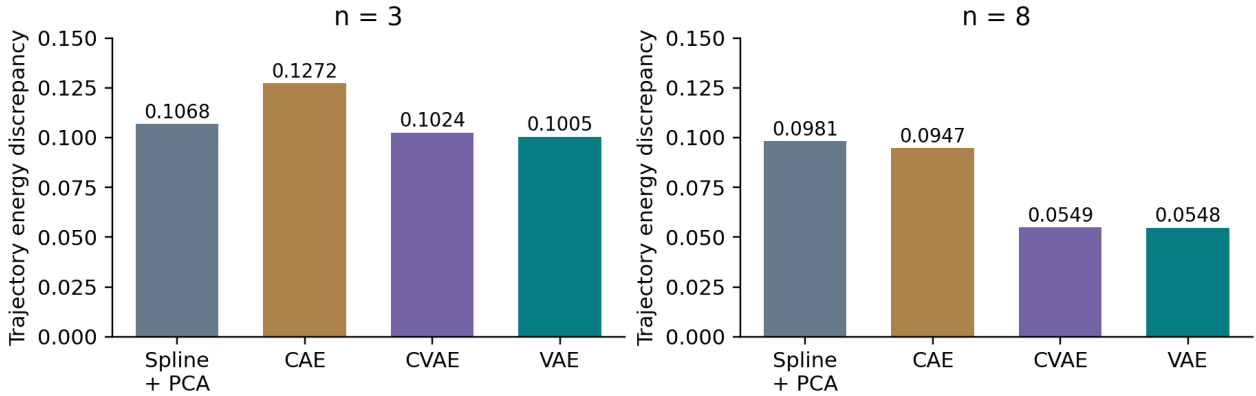


Figure 2. Cohort-average full-trajectory energy for all four compact models. The $n = 8$ VAE/CVAE advantages over spline survive both corrections and the axis-balanced sensitivity analysis (Appendix A8). At $n = 3$ their energy advantages are not significant, and their MMD advantages do not survive the axis-balanced BH analysis. Bars summarize 28 participants after seed averaging; they do not show uncertainty or a formal $n = 3$ versus $n = 8$ test.

An empirical reference that resampled a participant's stored context paths achieved mean energy 0.0328 and MMD 0.0054, below the compact models' means. This reference retains complete trajectories instead of an n -value center, so it addresses a different memory constraint, but it demonstrates that the compact generator does not yet recover all available session information. These reference differences are descriptive, without pairwise significance tests.

Repeating the context mean path gave a smaller mean-path RMSE (0.1146) than VAE $n = 8$ (0.1604), yet far worse energy (0.4621 versus 0.0548) and essentially zero generated variation. Thus distance to a mean path, or average error to all recorded paths, cannot alone establish successful stochastic generation. Distribution comparisons reward both agreement and an appropriate spread, rather than allowing an average-only output to appear sufficient.

7 Personal information and the dashboard

Good generation does not by itself show that the compact center carries information about its owner. We therefore held the decoder, task draws, covariance and random variation fixed, and replaced the center with either the training-population center or centers from other test participants. Using the participant's own context improved feature distributions for CVAE at both capacities and for VAE at eight dimensions (Figure 3). The VAE comparisons covered six tests and the CVAE comparisons twelve; every comparison favored the own-context center after BH correction. This supports useful personal information within a session, rather than stability across separate sessions.

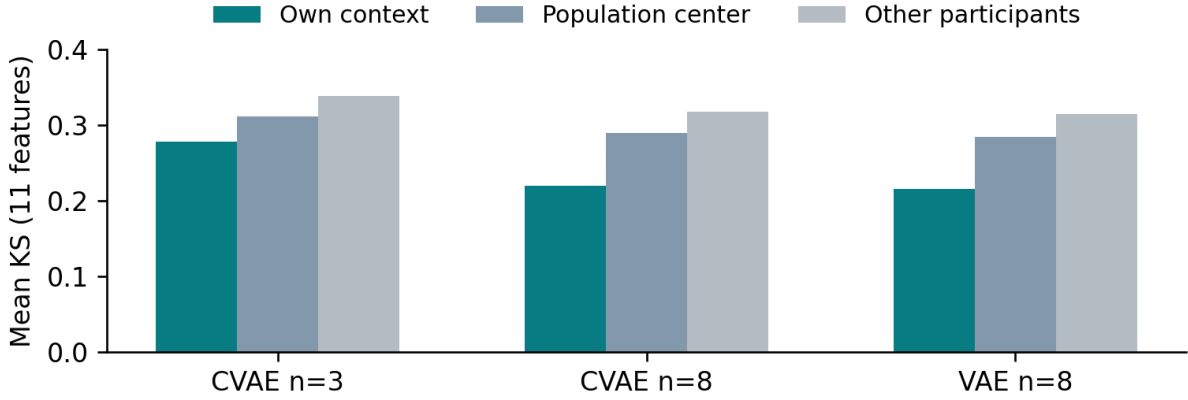


Figure 3. Mean KS under replacement of the latent center; lower is better. VAE $n = 8$ means were 0.2151 (own), 0.2841 (population) and 0.3144 (other participants). All six VAE comparisons across KS, energy and MMD had BH $q < 1.6 \times 10^{-6}$; all twelve CVAE comparisons had $q < 0.000382$. Control bars use saved study means. These feature controls do not establish that each latent coordinate has an interpretable meaning, and were not repeated for the new full-path endpoints.

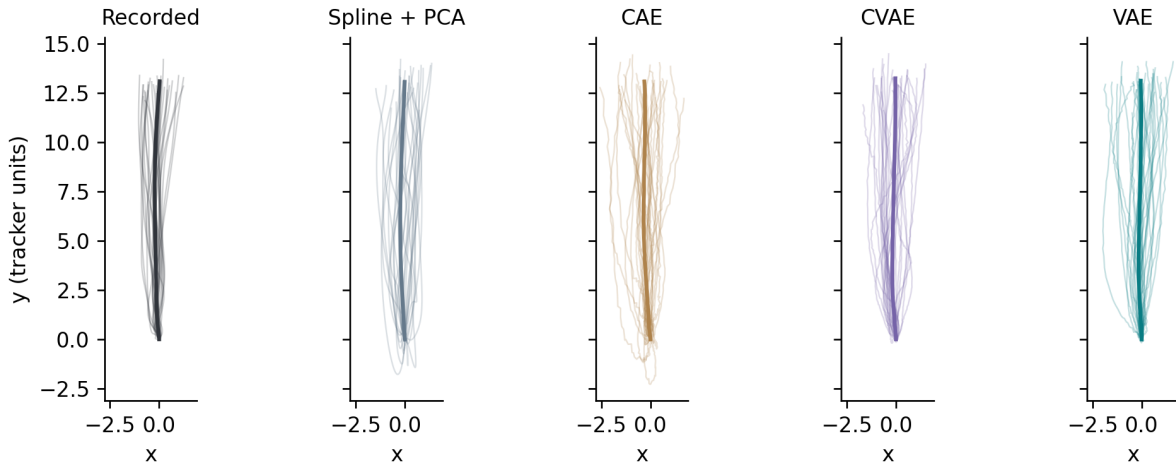


Figure 4. Recorded query and generated trajectories for subject01 at $n = 8$ (fold 2; neural seed 42). The first 30 stored paths illustrate spread; solid curves show means of all stored paths in each panel. Subject01 was chosen by identifier, without ranking model scores. Each panel uses equal x/y scale and shared limits. These examples illustrate outputs; inference uses all 28 participants. Individual generated paths have no one-to-one pairing with recorded trials.

The Streamlit dashboard integrates generation, visual exploration and benchmark comparisons. It supports CVAE and spline + PCA at $n = 2, 3, 4$ and 8, and CAE and VAE at $n = 3$ and 8. VAE $n = 8$ is the default because it combines the lowest mean feature discrepancies with the strong full-path results. CVAE remains a supported alternative, particularly when inspecting explicit task conditions. Spline + PCA provides the statistical comparison and strongest mean reconstruction. The VAE ignores condition controls; spline conditions affect timing only.

Live references use fold 0 and neural seed 42, while study score cards summarize the evaluated rotations and available seeds. The interface labels these scopes. The dashboard demonstrates the fitted models and does not supply independent validation of model selection. Appendix A9 gives reproducible launch instructions and a short demonstration sequence. No $n = 16$ model was evaluated in the matched study.

8 Discussion

The results separate faithful compression of an observed trajectory from useful generation of movement variability. Spline + PCA retained trajectory shape most accurately, while variational models at eight dimensions produced closer feature and whole-path distributions. This difference is consistent with the distinct tasks, but the experiments do not isolate a single causal explanation for the ranking. Our practical selection is VAE $n = 8$, with CVAE retained as a statistically comparable alternative for full-path generation. The original three-coordinate objective was only partly achieved: personal context was useful, but distribution fidelity and robustness were stronger at eight dimensions.

The evidence has several boundaries. Context and query trials came from the same recording session, and shared a task-condition mixture. Cross-session stability, unseen task extrapolation and a psychological interpretation of latent coordinates remain open. The noise covariance was shared across participants, and context averaging discarded information retained by empirical resampling. Direct context summaries also outperformed latent linear probes on the fourteen tested behavioral-summary targets (Appendix A10). We therefore regard the fingerprint as a useful representation for this decoder, rather than a sufficient representation of every measured individual difference.

Timing reconstruction is especially relevant to derivative-based features. Phase normalization does not encode seconds directly, and generated features depend on predicted durations and an onset-based crop. The event audit identified an arrival-proxy offset, zero-initiation trials and sensitivity to onset assumptions. Extreme timing predictions were retained in the evaluation, including an approximately 11.5-second initiation prediction in one CVAE $n = 8$ run. The full-path follow-up avoids the generated movement crop, but assesses paths in normalized phase and cannot validate physical timing by itself.

The comparisons are exploratory. Participants were held out from model fitting, but rotations shared training participants, three neural seeds do not exhaust training variability, and the dashboard default was chosen using the evaluated cohort. The $n = 3$ results depended on the distance metric and axis scaling. Nonsignificant VAE-CVAE differences do not show that task conditions are irrelevant: conditioning routes were active, and these architectures and settings are only one test of their value. A new cohort or prospectively held-out model-selection evaluation would strengthen the ranking. Feature reduction should be justified using the task and training data, with results reassessed under a specified protocol rather than selected to improve significance.

9 Conclusion

A compact context-derived representation can improve generation of a participant's interception-movement distribution, but reconstruction and generation favor different models. VAE and CVAE at eight dimensions provide the strongest compact full-path generation evidence relative to spline + PCA. The dashboard makes these models accessible, while the empirical resampling reference and unsuccessful summary probes identify information still lost through compression. Further work can examine richer participant distributions, condition-specific fidelity and generalization to new sessions.

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